

LANDFORMS OF GLACIATION / HIMALAYAN GLACIERS-GLOF

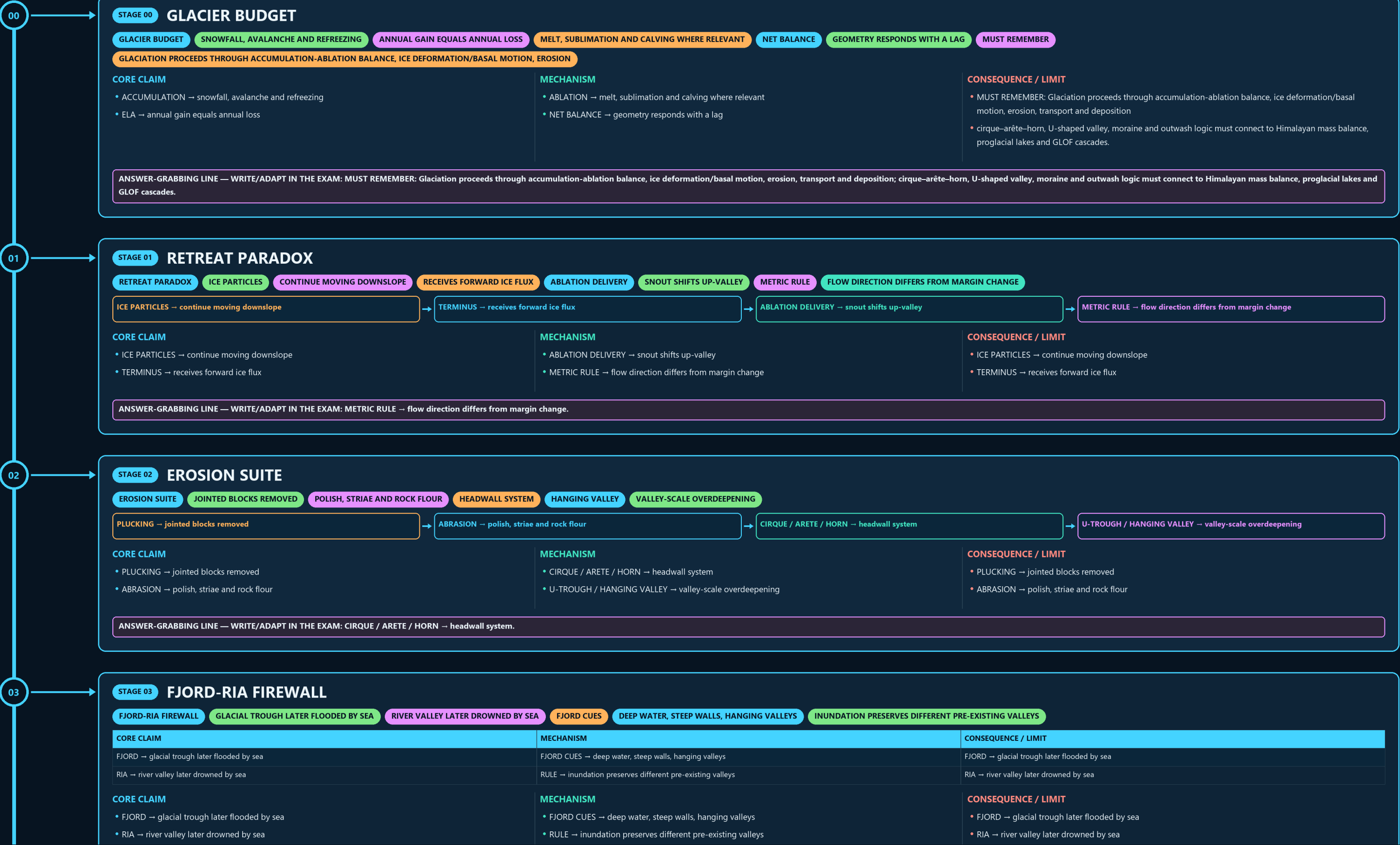
GLACIER BUDGET → RETREAT PARADOX → EROSION SUITE → FJORD-RIA FIREWALL → DEPOSITIONAL SORTING → ICE-MASS FAMILY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g6 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CIRQUE / ARETE / HORN → headwall system.

03

STAGE 03 FJORD-RIA FIREWALL

FJORD-RIA FIREWALL

GLACIAL TROUGH LATER FLOODED BY SEA

RIVER VALLEY LATER DROWNED BY SEA

FJORD CUES

DEEP WATER, STEEP WALLS, HANGING VALLEYS

INUNDATION PRESERVES DIFFERENT PRE-EXISTING VALLEYS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FJORD → glacial trough later flooded by sea	FJORD CUES → deep water, steep walls, hanging valleys	FJORD → glacial trough later flooded by sea
RIA → river valley later drowned by sea	RULE → inundation preserves different pre-existing valleys	RIA → river valley later drowned by sea
CORE CLAIM • FJORD → glacial trough later flooded by sea • RIA → river valley later drowned by sea	MECHANISM • FJORD CUES → deep water, steep walls, hanging valleys • RULE → inundation preserves different pre-existing valleys	CONSEQUENCE / LIMIT • FJORD → glacial trough later flooded by sea • RIA → river valley later drowned by sea

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FJORD CUES → deep water, steep walls, hanging valleys.

04

STAGE 04 DEPOSITIONAL SORTING

DEPOSITIONAL SORTING

DIRECT ICE, GENERALLY UNSORTED

MELTWERter RIDGE IN OR UNDER ICE

BRAIDED MELTWERter, RELATIVELY SORTED

STREAMLINED DRIFT/TILL LANDFORM

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
TILL / MORaine → direct ice, generally unsorted	OUTWASH → braided meltwater, relatively sorted	TILL / MORaine → direct ice, generally unsorted
ESKER → meltwater ridge in or under ice	DRUMLIN → streamlined drift/till landform	ESKER → meltwater ridge in or under ice
CORE CLAIM • TILL / MORaine → direct ice, generally unsorted • ESKER → meltwater ridge in or under ice	MECHANISM • OUTWASH → braided meltwater, relatively sorted • DRUMLIN → streamlined drift/till landform	CONSEQUENCE / LIMIT • TILL / MORaine → direct ice, generally unsorted • ESKER → meltwater ridge in or under ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESKER → meltwater ridge in or under ice.

05

STAGE 05 ICE-MASS FAMILY

ICE-MASS FAMILY

GLACIER, ICE CAP, ICE SHEET

SHELF, ICEBERG AND SEA ICE

SEA LEVEL

GROUNDING LOSS ADDS OCEAN MASS

SHELF LOSS CAN ACCELERATE INLAND DISCHARGE

GROUNDING → glacier, ice cap, ice sheet

FLOATING → shelf, iceberg and sea ice	SEA LEVEL → grounding loss adds ocean mass	BUTTRESSING → shelf loss can accelerate inland discharge
CORE CLAIM • GROUNDING → glacier, ice cap, ice sheet • FLOATING → shelf, iceberg and sea ice	MECHANISM • SEA LEVEL → grounding loss adds ocean mass • BUTTRESSING → shelf loss can accelerate inland discharge	CONSEQUENCE / LIMIT • GROUNDING → glacier, ice cap, ice sheet • FLOATING → shelf, iceberg and sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUTTRESSING → shelf loss can accelerate inland discharge.

06

STAGE 06 HIMALAYAN MAP HOOKS

HIMALAYAN MAP HOOKS

BARA SHIGRI

CHENAB SYSTEM

NUBRA-SHYOK-INDUS SYSTEM

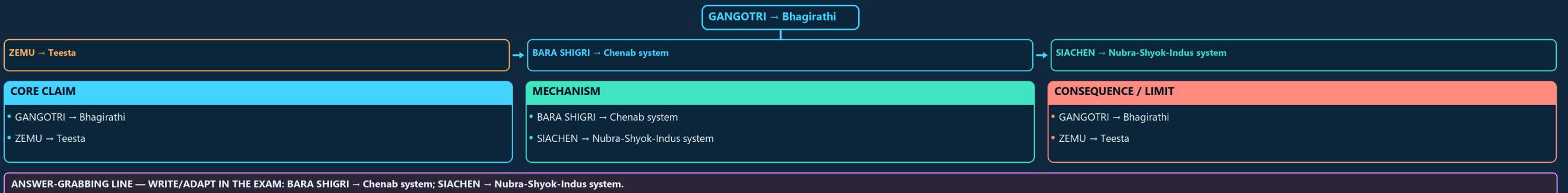
GANGOTRI → Bhagirathi

ZEMU → Teesta	BARA SHIGRI → Chenab system	SIACHEN → Nubra-Shyok-Indus system
CORE CLAIM • GANGOTRI → Bhagirathi • ZEMU → Teesta	MECHANISM • BARA SHIGRI → Chenab system • SIACHEN → Nubra-Shyok-Indus system	CONSEQUENCE / LIMIT • GANGOTRI → Bhagirathi • ZEMU → Teesta

06

STAGE 06 HIMALAYAN MAP HOOKS

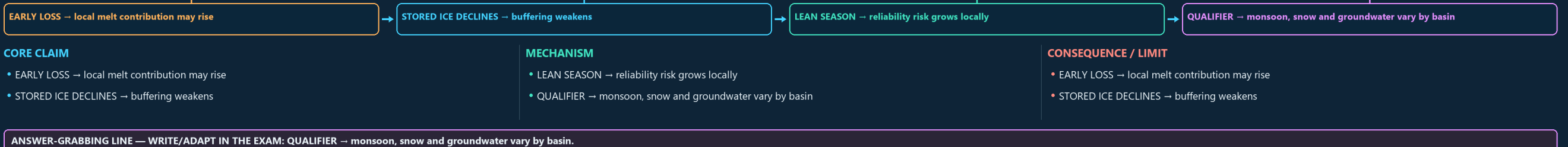
HIMALAYAN MAP HOOKS BARA SHIGRI CHENAB SYSTEM NUBRA-SHYOK-INDUS SYSTEM



07

STAGE 07 WATER-TIMING SEQUENCE

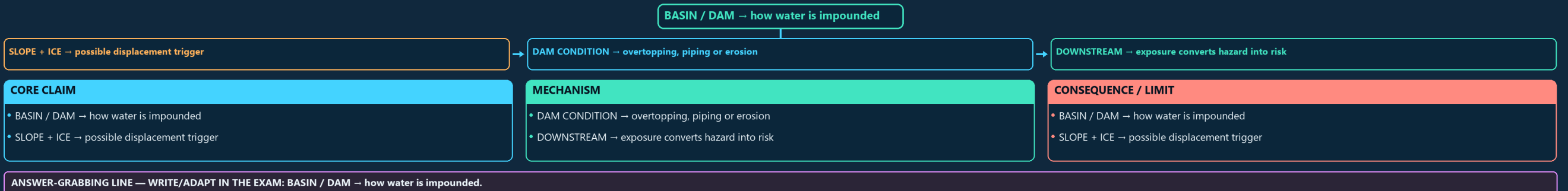
WATER-TIMING SEQUENCE EARLY LOSS LOCAL MELT CONTRIBUTION MAY RISE STORED ICE DECLINES BUFFERING WEAKENS LEAN SEASON RELIABILITY RISK GROWS LOCALLY MONSOON, SNOW AND GROUNDWATER VARY BY BASIN



08

STAGE 08 GLACIAL-LAKE TEST

GLACIAL-LAKE TEST HOW WATER IS IMPOUNDED SLOPE + ICE POSSIBLE DISPLACEMENT TRIGGER DAM CONDITION OVERTOPPING, PIPING OR EROSION EXPOSURE CONVERTS HAZARD INTO RISK



09

STAGE 09 GLOF CASCADE

GLOF CASCADE SLOPE FAILURE, INFLOW OR INSTABILITY DAM EROSION BEGINS SUDDEN RELEASE + SEDIMENT CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES CLOSE DISTINCTION SNOWLINE, EQUILIBRIUM-LINE ALTITUDE, FIRN LINE AND GLACIER TERMINUS ARE

